UNIVERSITY OF NEW SOUTH WALES

ACTL4001: Actuarial Theory & Practice A

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Assessment 2: Group Project

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1. Executive Summary

The Galaxy General Insurance Company recommends accepting the Cosmic Quarry Mining Corporation RFP and proceeding with all four product lines at various commitments. In our actuarial modelling across 454,914 historical policy records, it is demonstrated that each line is priceable, commercially viable and produces a well-diversified portfolio. Our general recommendation for the four product lines is:

- Equipment failure: a mature product line with an expected annual aggregate loss of $88.8m, coefficient of variation of 4.3% and gross premium of $129.9m = straightforward to price
- Cargo loss: the highest volume and highest aggregate line product with expected loss of $32.1 trillion and gross premium of $45.9 trillion driven by shipment scale and route hazard = proven to be profitable within disciplined underwriting.
- Workers' compensation: low frequency (0.028 per exposure year) and low individual claim cost, but long-tail duration risk. The expected aggregate was $58.9k with a gross premium of $86.8k = simple to administer and necessary for Cosmic Quarry ethics.
- Business interruption: the most volatile product line with CV of 10.7%, $\text{TVaR}_{99.5}$ reaching $162.5m against an expected $123.8m = we recommend this under mandatory waiting periods, limits and minimum backup standards particularly in Oryn Delta.

All gross premiums include a 12% expense loading, 8% profit margin, and a 10% capital charge on the technical premium. Long-term net present values are positive across all lines at a 3.76% discount rate, confirming the portfolio's financial attractiveness over a multi-year horizon.

The most significant risk that poses to these product lines is a correlated solar storm which can affect all three solar systems simultaneously which can be captured by our worst-case scenario testing which estimates effects lifting total portfolio costs to over $57 trillion which is ~78% above our base case of $32.1 trillion. This heavy tailed risk can be both managed and prepared for through a suggestion of cross system reinsurance, per occurrence limits and structured extensions of particular coverages. Model risk from the imperfect mapping of historical data to the new Cosmic Quarry solar systems is another major concern which ultimately justifies the conservative risk margins embedded in our pricing. Additionally, as the Cosmic Quarry expands in equipment and routes we believe our coverages can be adjusted to scale suitably.

Across all four hazard areas, the proposed product designs are intentionally structured to align insurance pricing with operational decision-making, thereby embedding risk awareness into Cosmic Quarry Mining Corporation's enterprise risk management framework.

Hazard Area	ERM Integration Recommendation
Equipment Failure	Integrate asset-condition data with the policy trigger; age-based deductible schedules create direct incentive alignment with Cosmic Quarry's preventive maintenance programme.
Cargo Loss	Route-risk premiums make interplanetary trade-off decisions visible at the business unit level; fleet-level cargo limits support risk aggregation reporting to the Board.
Workers' Comp.	Role and gravity-adjusted pricing incentivises deployment of experienced staff to high-risk stations; experience-rating credits reward safety-training investment over time.
Business Interruption	Waiting periods and redundancy conditions push operational resilience standards into Cosmic Quarry's procurement and infrastructure decisions, aligning insurance with ERM governance.

Table 1 – ERM integration recommendation strategy as per hazard area

2. Product Design

2.1 Equipment Failure

The equipment failure coverage is provided on a repair and replacement cost basis for sudden and accidental mechanical or operational failure of insured equipment. This policy covers the cost of repair or replacement when the equipment is beyond economic repair. This policy is strategically recommended due to its low severity resulting in relatively predictable losses, alongside rich historical data allowing for strong credibility in our models. Moreover, equipment failure is directly tied to the core mining operation and often has lower correlations to the catastrophic system wide effects. Thus, we believe that the benefits should be structured around three tiers.

- Tier 1: complete replacement cost for equipment under 3 years old and within active maintenance compliance, subject ot policy limit
- Tier 2: repair cost and an additional depre4ciated replacement contribution for equipment between 3 to 7 years old, with depreciation linked to the maintenance compliance index.
- Tier 3: repair cost only for equipment over 7 years old with minimum deductible of 15% of the repair estimation cost.

Downtime costs (loss of production output during repair) are covered up to a sublimit of 20% of the repair or replacement benefit, subject to a 48-hour waiting period. This provides meaningful protection against incidental production loss without duplicating the Business Interruption product.

The policy is triggered by sudden, unforeseen mechanical or operational failures which results in physical damage to the insured equipment rendering the equipment inoperable or have failure downtimes exceeding the applicable waiting period. Additionally, the failure must be attributable to a covered peril including material fatigue, operational overload within rated capacity, or environmental stress and the claim must be during the period of insurance. The failure must be confirmed by an independent field inspection report submitted within 10 days of the incident.

The following are excluded unless specifically endorsed:

- Wear, tear, corrosion or gradual deterioration, and failure resulting from non-compliance with the insured's scheduled maintenance program.
- Intentional misuse, operator error caused by insufficient certified training or unapproved modifications to the equipment.
- Cyber-induced failure or sabotage unless a cyber endorsement is purchased.
- Consequential or indirect losses beyond the equipment itself, covered instead under the Business Interruption product.

As Cosmic Quarry expands its equipment fleet or introduces new equipment types, the tier structure can be extended without renegotiating the core policy terms. The Cyber Endorsement provides a natural pathway to cover emerging technology risks as digitally integrated equipment becomes more prevalent across the fleet.

2.2 Cargo Loss

This policy covers the physical loss or damage to insured cargo during interstellar transit between the solar systems based on their declared value of the cargo. Maximum limits are set as a multiple of declared insured values with hard limits applied to specific route specific risk classes. Shipments which exceed this maximum limit will require specific agreements of terms. The benefit limits will be set by route class:

- Route class 1: low debris density and low solar radiation (index < 0.3), this coverage class will allow up to 100% of the declared value with a standard deductible of 0.5%
- Route class 2: moderate hazard routes with solar radiation indexes between 0.3 and 0.6, allowing for up to 90% of the declared value with a deductible of 1%
- Route class 3: refers to any high hazard routes with indices above 0.6 and / or transit durations longer than 3 years (36 months) which will only allow up to 80% of the declared value and deductible of 2.5%

Based on historic data, the cargo loss distribution is typically heavily right-skewed, this can be seen in our lognormal severity model with a mean severity of approximately 7.8 million currency units against a maximum observed claim of 678 million making robust limit structures essential to managing tail exposure.

The policy is triggered by physical loss or damage to insured cargo occurring during the insured transit timeline. Evidence of loss must be presented via a manifest discrepancy, damage survey, or vessel incident report.

The following are excluded unless specifically endorsed:

- Unmanifested shrinkage, mysterious disappearance, or inventory shortfall without physical evidence of loss.
- Ordinary leakage, normal weight loss.
- Any contraband, prohibited goods, or cargo transported on unlicensed or sanctioned routes.
- Loss arising from routes designated as prohibited under the applicable route registry unless an endorsed high-risk lane extension is purchased.
- Loss or damage to goods caused by the misconduct of the insured or its agents.

The route-class framework is designed to accommodate new transit corridors as the interstellar trade network expands. New destinations can be added to the approved route schedule by endorsement, with pricing derived directly from the debris density and solar radiation indices for the new lane. This avoids the need for full policy re-issuance as Cosmic Quarry's trade network grows.

2.3 Workers' Compensation

This policy provides medical expense coverage and wage replacement for workers who sustain a work related injury (physical and psychological) during the coverage period. Three standard benefits apply:

- Medical reimbursement: reasonable and necessary treatment cost for the covered injury including hospitalization, rehabilitation and checkups / physio therapy.
- Wage replacement: indemnity benefits up to an equal to 75% of the worker's base salary pro-rated for the duration of the certified incapacity, subject to a maximum benefit period of 1,000 days.
- Lump sum permanent impairment benefits: a lump sump payout for injuries resulting in permanent partial or total disability calculated as a multiple of base annual salary.

Role-based pricing is applied at inception using occupation class, gravity level, and hours-per-week bands. A Safety Training Credit of up to 6% premium reduction is available for stations achieving a safety training index score of 4 or 5. Psychological claims are reflected in the significance of the psych_stress_index variable in our frequency model and the prevalence of psychological claims in the historical data.

Cover is triggered by a work-related injury (physical and psychological) because of and during covered employment within the policy period. A medical report must be filed within 30 days of the incident to initiate any claim.

The following are excluded unless specifically endorsed:

- Pre-existing conditions not aggravated by covered employment.
- Self inflicted injuries or those arising from under the influence
- Independent contractors and casual labour not listed on the covered schedule.
- Injuries sustained during activities expressly outside the scope of covered employment.
- Claims without independent clinical certification.

The role-based rating structure and gravity/fatigue adjustments are parameterised as multiplicative relativity factors, making them straightforward to extend as Cosmic Quarry hires into new occupational categories or enters environments with different gravitational profiles. Safety-incentive credits can be linked to a verifiable annual audit, creating a dynamic pricing mechanism that rewards continued investment in workforce safety.

2.4 Business Interruption

The policy covers loss of gross revenue and necessary extra expenses incurred as a direct result of an interruption to mining operations caused by a covered event. Cover is structured with explicit financial controls to contain the high aggregate volatility observed in our modelling (TVaR99.5 of $165.2M against an expected $123.8M):

- Revenue loss: indemnity for revenue losses during the period calculated as the difference between actual revenue and interrupted revenue, based on a 12 month rolling average.
- Extra expense: additional costs to restore operations such as temporary equipment hire, accelerated maintenance with a limit up to 30% of the revenue loss.
- Maximum indemnity period: 12 months per occurrence with a 72 hour waiting period before indemnity commences with extended indemnity periods of up to 24 months available.

The coverage is triggered by operational interruptions at certain mining stations caused by covered equipment failure, logistics breakdowns on supply routes or power / energy failure at the station with the interruption resulting in a measurable reduction in production output of at least 20% relative to the 30 day pre event average, confirmed by independent records.

The following are excluded unless specifically endorsed:

- Non damage market loss, due to commodity prices, trade sanctions or demand side factors.
- Regulatory or government ordered shutdowns unless caused by covered equipment.
- Unknown interruptions with unresolved defects in equipment or infrastructure.
- Losses during waiting period or exceeding the per system annual aggregate limit.

All products were designed to be tailored to the distinct operating environment of each solar system. table 6 in the appendix covers the key product adaptations across solar systems. The Bayesia system serves as the pricing benchmark given its balanced risk profile while Helionis receives credits for its greater redundancy and operational length. Finally, Oryn Delta attracts higher loadings, tighter wordings and stricture conditions reflecting its greater operational fragility and lower resilience infrastructure.

3. Pricing & Capital Modelling

In conducting an actuarial assessment of Cosmic Quarry Mining Corporation's pricing adequacy and capital requirements for the proposed insurance program, the Galaxy General Insurance Company's follows best practice for multi-line insurance portfolios under complex risk environments with three key components. First, independent claim severity and frequency severity modelling across each hazard class using Generalised Linear Models (GLMs), incorporating relevant operational and environmental risk drivers. Second, aggregate losses were simulated using Monte Carlo techniques, combining stochastic claim counts with simulated claim severities to produce full empirical loss distributions for each line of business. Third, pricing and capital metrics were derived from these simulated distributions, including expected losses, variances, and tail measures.

3.1 Aggregate Loss Distributions by Hazard Coverage

The aggregate loss distributions form the foundation of both pricing and capital assessment by outlining the risk profile of each hazard as well as the entire portfolio. Claim frequency and severity for each hazard were modelled using GLMs, incorporating key operational drivers associated with each hazard such as equipment usage, environmental factors, and workforce characteristics.

Line of Business	Expected Loss	Standard Deviation	VaR (99%)	TVaR (99%)	VaR (99%) / Expected Loss
Equipment Failure	8.88E+07	3.85E+06	9.81E+07	9.93E+07	110%
Cargo loss	3.21E+16	5.53E+12	3.21E+16	3.21E+16	100%
Workers' Compensation	5.89E+04	3.31E+03	6.67E+04	6.80E+04	113%
Business Interruption	1.24E+08	1.33E+07	1.57E+08	1.62E+08	127%
Total Portfolio	3.21E+16	5.53E+12	3.21E+16	3.21E+16	100%

Table 2 - Aggregate Loss Distributions by Hazard Coverage

The aggregate loss distributions in Table 2 reveal a highly asymmetric and concentrated portfolio structure. Cargo loss dominates the portfolio in absolute terms, with expected losses far exceeding that of any other hazard class. Despite its extremely low relative volatility, this scale results in cargo effectively wholly determining the total portfolio distribution.

Business interruption exhibits the highest relative volatility and the most pronounced right tail. The 99th percentile loss is approximately 27% higher than the mean, indicating a significant sensitivity to extreme operational disruption and identifies business interruption as the primary driver of tail risk from a relative risk standpoint. Whilst the 99^{th} percentile loss is equivalent to the mean for cargo loss, it will still drive portfolio-level tail risk due to its magnitude. Equipment failure and workers' compensation display comparatively stable distributions. Their limited variance and modest tail behaviour suggest that these lines are largely attritional and operationally controllable, contributing positively to earnings without materially affecting capital requirements.

3.2 Pricing Analysis & Profitability

Pricing is based on Monte Carlo simulations of aggregate losses combining stochastic claim counts with simulated claim severities, generating empirical distributions for each hazard and the overall portfolio. The expected costs represent the actuarially fair premium and additional loadings incorporated to reflect expenses, profit and capital requirements. Gross premiums include a 12% expense loading, 8% profit margin and 10% capital charge, ensuring that pricing remains both commercially viable and actuarially sufficient to absorb volatility and tail risk.

3.2.a Short Term Ranges: Costs, Returns and Net Revenue

Short-term economic outcomes are derived from the simulated annual distributions of aggregate losses, premiums, and net revenue. These results provide a distributional view of performance, capturing not only expected values but also variability and extreme outcomes.

Line of Business	Metric	Mean	Variance	P5	P50	P95	P99	TVaR (99%)
Equipment failure	Cost	8.88E+07	1.48E+13	8.26E+07	8.87E+07	9.53E+07	9.81E+07	9.93E+07
	Return	1.30E+08	0.00E+00	1.30E+08	1.30E+08	1.30E+08	1.30E+08	1.30E+08
	Net Revenue	4.11E+07	1.48E+13	3.46E+07	4.12E+07	4.73E+07	4.98E+07	5.10E+07
Cargo loss	Cost	3.21E+16	3.06E+25	3.21E+16	3.21E+16	3.21E+16	3.21E+16	3.21E+16
	Return	4.59E+16	0.00E+00	4.59E+16	4.59E+16	4.59E+16	4.59E+16	4.59E+16
	Net Revenue	1.38E+16	3.06E+25	1.38E+16	1.38E+16	1.38E+16	1.38E+16	1.38E+16
Workers' compensation	Cost	5.89E+04	1.10E+07	5.35E+04	5.89E+04	6.44E+04	6.67E+04	6.80E+04
	Return	8.68E+04	0.00E+00	8.68E+04	8.68E+04	8.68E+04	8.68E+04	8.68E+04
	Net Revenue	2.78E+04	1.10E+07	2.24E+04	2.79E+04	3.33E+04	3.55E+04	3.65E+04
Business interruption	Cost	1.24E+08	1.77E+14	1.03E+08	1.24E+08	1.46E+08	1.57E+08	1.62E+08
	Return	1.88E+08	0.00E+00	1.88E+08	1.88E+08	1.88E+08	1.88E+08	1.88E+08
	Net Revenue	6.40E+07	1.77E+14	4.15E+07	6.41E+07	8.51E+07	9.33E+07	9.72E+07
Total portfolio	Cost	3.21E+16	3.06E+25	3.21E+16	3.21E+16	3.21E+16	3.21E+16	3.21E+16
	Return	4.59E+16	0.00E+00	4.59E+16	4.59E+16	4.59E+16	4.59E+16	4.59E+16
	Net Revenue	1.38E+16	3.06E+25	1.38E+16	1.38E+16	1.38E+16	1.38E+16	1.38E+16

Table 3 - Short Term (1 Year) Economic Ranges by Line of Business

The distributional view of annual outcomes for each hazard class in Table 3 reveals that all four hazard areas generate positive expected net revenue in the short-term, confirming overall pricing adequacy, although with a highly uneven distribution of profitability across the portfolio. Equipment failure and workers' compensation produce stable and predictable earnings due to low variance and narrow distributional ranges. However, as expected, cargo loss drives the majority of absolute profit due to its scale relative to the portfolio, while business interruption contributes meaningful but highly volatile returns due to its heavy-tailed loss distribution. While expected net revenue is positive for all lines and shows minimal variance in the short-term, the extreme concentration in one line of business underscores the importance of incorporating tail-based risk margins and ensuring that pricing is supported by adequate capital buffers.

3.2.b Long Term Ranges: Costs, Returns and Net Revenue

The long-term analysis extends the short-term framework by considering the present value of future costs, returns and net revenue over the projection horizon (10 years).

Line of Business	Metric	Mean	Variance	P5	P50	P95	P99	TVaR (99%)
Equipment failure	PV Cost	8.09E+08	1.21E+14	1.10E+07	7.72E+08	7.84E+08	7.91E+08	8.09E+08
	PV Return	1.21E+09	0.00E+00	0.00E+00	1.21E+09	1.21E+09	1.21E+09	1.21E+09
	PV Net Revenue	4.02E+08	1.21E+14	1.10E+07	3.66E+08	3.76E+08	3.83E+08	4.02E+08
Cargo loss	PV Cost	2.93E+17	2.63E+26	1.62E+13	2.93E+17	2.93E+17	2.93E+17	2.93E+17
	PV Return	4.28E+17	0.00E+00	0.00E+00	4.28E+17	4.28E+17	4.28E+17	4.28E+17
	PV Net Revenue	1.35E+17	2.63E+26	1.62E+13	1.35E+17	1.35E+17	1.35E+17	1.35E+17
Workers' compensation	PV Cost	5.37E+05	9.25E+07	9.62E+03	5.02E+05	5.16E+05	5.22E+05	5.37E+05
	PV Return	8.09E+05	0.00E+00	0.00E+00	8.09E+05	8.09E+05	8.09E+05	8.09E+05
	PV Net Revenue	2.72E+05	9.25E+07	9.62E+03	2.41E+05	2.49E+05	2.56E+05	2.72E+05
Business interruption	PV Cost	1.13E+09	1.48E+15	3.84E+07	9.81E+08	1.03E+09	1.07E+09	1.13E+09
	PV Return	1.75E+09	0.00E+00	0.00E+00	1.75E+09	1.75E+09	1.75E+09	1.75E+09
	PV Net Revenue	6.24E+08	1.48E+15	3.84E+07	4.85E+08	5.34E+08	5.59E+08	6.24E+08
Total portfolio	PV Cost	2.93E+17	2.58E+26	1.61E+13	2.93E+17	2.93E+17	2.93E+17	2.93E+17
	PV Return	4.28E+17	0.00E+00	0.00E+00	4.28E+17	4.28E+17	4.28E+17	4.28E+17
	PV Net Revenue	1.35E+17	2.58E+26	1.61E+13	1.35E+17	1.35E+17	1.35E+17	1.35E+17

Table 4 - Long Term (10-year PV) Economic Ranges by Line of Business

The long-term results in Table 4 demonstrate that all lines remain profitable on a present value basis, with positive expected net revenue sustained across the projection period of 10 years. However, variability increases materially over time due to the compounding effect of uncertainty. For cargo loss, the present value of aggregate costs is approximately $2.93 × 10^{17}, with a wide distribution reflecting cumulative exposure. The corresponding net revenue of approximately $1.35 × 10^{17}, almost the entirety of the portfolio PV Net Revenue, confirms the dominant contribution of this line to long-term profitability. Business interruption shows a present value cost of approximately $1.13 billion, with significant dispersion across percentiles, reflecting the persistence of tail risk over time. The variability of long-term outcomes reinforces the need for conservative capital management, particularly in environments where correlated disruptions may occur. Equipment failure and workers' compensation maintain stable long-term profiles, with moderate variability and consistently positive net revenue, providing a reliable earnings base that partially offsets volatility from other lines. Overall, the long-term distributions confirm that while expected profitability is strong, the increased time uncertainty and concentration in cargo makes it imperative for capital requirements to account for the potential accumulation of adverse outcomes.

3.3 Stress Testing & Extreme Scenarios

To evaluate the portfolio's sensitivity to extreme scenarios, a structured scenario-based methodology was implemented to assess the resilience of the portfolio under severe but plausible conditions. Building on the Monte Carlo simulations, the stress testing framework additionally introduces simultaneously introduces systematic shifts to key risk drivers across all hazard classes, capturing incremental portfolio stress and somewhat capturing correlated extreme events (although not explicitly modelled using Copulas) that are not fully reflected in independent modelling assumptions.

To reflect the impact of plausible events such as large-scale solar storms, logistic breakdowns and infrastructure failures across multiple solar systems, stress scenarios were constructed by applying adverse shocks to frequency and severity parameters across all lines of business. These shocks were calibrated to approximate a 1-in-100-year event as the worst-case scenario, consistent with standard actuarial capital modelling practices. The resulting stressed loss distributions were then re-simulated to generate portfolio-level outcomes under each scenario.

Scenario	Expected Cost	Standard Deviation	VaR (95%)	VaR (99%)	VaR (99.5%)	TVaR (95%)	TVaR (99%)	TVaR (99.5%)
Best case	2.87E+16	8.88E+12	2.87E+16	2.87E+16	2.87E+16	2.87E+16	2.87E+16	2.87E+16
Moderate / base case	3.21E+16	5.53E+12	3.21E+16	3.21E+16	3.21E+16	3.21E+16	3.21E+16	3.21E+16
Worst case	5.72E+16	8.10E+12	5.72E+16	5.72E+16	5.72E+16	5.72E+16	5.72E+16	5.73E+16

Table 5 - Scenario testing summary

The results of this analysis demonstrate a significant increase in aggregate losses under extreme conditions. In the base case, total portfolio costs are approximately $32.1 trillion, while under the worst-case scenario, costs increase to approximately $57 trillion, representing an increase of approximately 78%. This uplift reflects the combined impact of increased claim frequency, higher severity, and high cargo concentration of the portfolio, which importantly highlights the asymmetry of risk within the portfolio. While favourable scenarios result in relatively modest reductions in losses, adverse scenarios produce disproportionately large increases, particularly when correlated solar system events may have a tendency to impact individual lines of businesses. This indicates that the portfolio is exposed to downside risk that is not fully captured by expected values or even standard percentile measures.

From a capital perspective, the implications of these findings are that a reliance on mean-based pricing or even standalone VaR measures may be detrimental to Cosmic Quarry Mining Corporation. Instead, capital requirements should be calibrated using stressed scenarios and tail metrics such as TVaR, ensuring that the portfolio remains solvent under extreme but plausible conditions.

3.4 Implications & Strategic Recommendations

The pricing and capital modelling results collectively demonstrate that the proposed portfolio is profitable and scalable, but is structurally exposed to concentration and tail risk. From a strategic perspective, this implies that pricing discipline and risk management must be closely aligned. Cargo exposures require strict underwriting controls and aggregation limits to manage concentration risk. Business interruption coverage must be carefully structured with waiting periods, limits, and operational requirements to mitigate tail exposure. Furthermore, capital allocation should be driven by stress scenarios rather than expected outcomes, ensuring that sufficient buffers are maintained to absorb extreme losses. The integration of pricing with operational risk drivers (e.g. route risk, equipment condition, and workforce characteristics) also supports alignment with Cosmic Quarry's enterprise risk management framework.

4. Risk Assessment

The key drivers of equipment failure frequently was equipment age, usage intensity and maintenance interval. The historical data which observed a frequency of 0.157 claims per exposure year revealed that older more intensively operated machines with longer intervals between maintenance cycles are materially more likely to fail with severity ranges from 11k to 790k demonstrating a log-normal distribution. The Helionis cluster held the greatest concentration of equipment and consequently the largest aggregate exposure, whilst Oryn Delta presented the highest equipment tail risk despite its smaller and less resilient setup. Cargo loss' claim frequency is highest amongst all the lines at 0.49 claims per exposure year which is driven by several case specific factors. The severity is highly skewed with the mean of 7.8 million greatly exceeding the median of 381,000 reflecting occasional catastrophic losses up to 678m. Moreover, the log normal distribution was selected as the best fitting severity model where termination of routes in or departing from Oryn Delta resulted in higher debris density. Workers compensation was the lowest in frequencies with 0.028 claims per exposure year across 134,947 workers. Benefit amounts were low but long-tail risk through claim length was contributed through hours worked and psychological stress indexes. Finally, business interruption had the widest aggregate loss range from 75m to 175m at a portfolio level primarily driven by production load, energy backup adequacy and supply chain concentration. Severity is clustered around 1.43m, indicated that most triggered claims almost reached maximum payout, confirmed by our Weibull severity model enforcing careful limit and waiting period design.

The most financially significant risks are those of high correlation primarily being a severe interstellar solar storm causing equipment failures, disruption in cargo transit, extended business interruption events and generating spikes in workers compensation claims through radiation exposure. Any supply chain cascades to a specific route will cause a correlated increase in claims across the network within a short network. Our aggregate modelling under the worst case correlated scenario produced a portfolio mean of 75 trillion currency units with a $TVaR_{99.5}$ of 57.25 trillion unit representing a 78% premium above base case expected costs, ultimately highlighting the importance of cross solar system reinsurance.

In our scenario testing, our base case reflected smooth operations with attritional claims only, no route disruptions and major events, resulting in expected portfolio costs of 28.7 trillion, a $TVaR_{99.5}$ of 28.7 trillion and a narrowing spread between the mean and tail metrics indicating that in absence of correlated events, the portfolio is well positioned and amenable to efficient pricing. Our moderate case represents the best estimate using fitted frequency and severity models across lines with general dependence assumptions. The expected portfolio costs were higher at 32.1 trillion and a $TVaR_{99.5}$ of 32.1 trillion. Our worst case scenario simulates catastrophic correlated events, effecting all solar systems across our main coverage areas, the expected portfolio costs was almost double of the base case with 57.2 trillion and a $TVaR_{99.5}$ of 57.3 trillion representing a almost 0.1% increase in spread.

Our threat table concluded correlated multi solar system disruptions causing the highest severity followed by equipment ageing, the operational fragility of Oryn Delta, cargo route debris and supply chain interruptions, each of this will medium to high likelihoods and effecting few lines of business (LoB). Further information can be found in table 7 outlining the entire threat table.

The Bayesian system serves as our pricing anchor with the Helionis cluster policies carrying concentration related loadings for the BI and equipment product lines. Policies for Oryn Delta require the most conservative terms across all the product lines with wider deductibles, stricter per-coverage inspections and higher limits on cargo.

5. Model Assumptions

Our pricing and capital results are derived from structure frameworks, where claim counts and severities are modelled separately for each hazard using generalized linear models, incorporating key operational and environmental risk drivers. These are then combined through a Monte Carlo simulation which repeatedly samples from fitted frequency and severity distributions to produce an empirical aggregate loss distribution for each product line and the total portfolio. Gross premiums are built from simulated expected losses, supplemented by TVaR based risk margins with fixed loadings of 12% ,8%, 10% for expenses, profit and capital respectively. Furthermore, frequency and severity are assumed to be conditionally independent provided the selected risk factors allowing them to be modelled and estimated separately. Claim frequency was modelled using a Poisson distribution, severity was modelled using a Lognormal distribution due to its right skew used to capture heavy tailed nature of losses. Most of our model parameters were treated as fixed rather than stochastic resulting in parameter uncertainty not being explicitly displayed through to the outputs. Inflation, discount rates and scale of operations were all assumed over the projection horizon and historical data is not representative of future experiences.

One important note would be that risks across hazard lines and solar systems are assumed independent in our base model with correlation instead captured through stress testing, this may result in dependency where baseline results could understate tail risk under certain systemic events. The results should therefore be interpreted alongside the scenario outcomes in our risk assessment which explains our conservative view on extreme loss potential.

6. Data Limitations and Impact

We did not utilise any other data sources than the one provided by the case study. Main limitations to the data came during the cleaning process, finding variables outside of the data dictionary ranges, missing data values and certain files not being split by solar system or stations. Table 8 covers the seven material data limitations in depth identified during our actuarial analysis. However, it is important to note that these limitations do not undermine the soundness and accuracy of our actuarial approach, but they do justify the conservative margins embedded in our pricing and the staged product rollout recommended for Oryn Delta.

In accordance with the case study / main assignment rules, our team discloses the following AI usage. Claude and ChatGPT was used to assist with drafting report sections, identifying key topics of discussion. However, the writing of the report has been completed solely by the members of the team and without the automation of AI. All quantitative modelling was conducted independently by the team in R, with outputs validated by manual cross-checks and reviewed by the team's actuarial lead. No substantive analytical conclusions were generated by AI without independent validation. Any AI assisted text was meticulously reviewed, revised, and approved by the team prior to submission.

Appendix

Line of Business	Helionis Cluster	Bayesia System	Oryn Delta
Equipment Failure	Premium credits for Helionis backup depth and redundancy standards; standard inspection terms.	Base-case terms; benchmark for pricing and maintenance inspection frequency.	Higher deductibles and mandatory inspection conditions; limited credits until track record established.
Cargo Loss	Standard route-class pricing; established convoy infrastructure supports lower loadings.	Base-case terms; route risk loadings applied at standard debris and radiation indices.	Stricter convoy requirements; higher debris-zone loadings; sublimits on high-risk lanes.
Workers' Compensation	Standard gravity and fatigue adjustments; lower contractor mix loading given established workforce.	Base-case adjustments; benchmark gravity index at near-Earth levels; standard safety credits.	Stronger gravity and contractor-mix loadings; elevated protective-gear requirements; tighter psych-stress provisions.
Business Interruption	Shorter waiting period (72 hrs) reflects stronger redundancy; higher sublimits on account of scale.	Standard 120-hr waiting period; moderate sublimits; base restoration clause terms.	Extended 168-hr waiting period; tighter sublimits; stricter minimum backup-power requirements.

Table 6 - Specific solar system adaptations

Threat	Affected LOB(s)	Likelihood	Severity	Rank	Mitigation
Correlated solar storm disrupting transport and operations	Cargo, BI, Equipment, WC	Medium	Extreme	1	Cross-system sublimits, reinsurance, coordinated emergency protocols
Equipment ageing concentration in core fleet	Equipment	High	High	2	Age-based maintenance warranties and inspection conditions
Oryn Delta operational fragility and lower redundancy	BI, Equipment, WC	Medium	High	3	System-specific deductibles, stronger backup standards, staged rollout
Cargo route debris surge	Cargo	High	High	4	Route-based pricing, convoy requirements, high-risk lane exclusions
Supply-chain interruption causing BI clustering	BI	Medium	High	5	Waiting periods, restoration clauses, minimum redundancy requirements
Labour injury accumulation from technical roles	WC	Medium	Medium	6	Role-based safety incentives and fatigue management protocols
Inflation shock on replacement and medical costs	All	Medium	High	7	Inflation indexing and annual repricing clauses
Model risk from imperfect historical-to-new solar system mapping	All	High	High	8	Conservative margin, scenario overlays, governance sign-off

Table 7 - Threat table

Limitation	Impact on Recommendations
Historical solar-system labels do not align with new business solar systems.	Requires relativity-based rather than pure credibility pricing by solar system; introduces model uncertainty.
No prospective BI exposure file exists at station level.	BI pricing relies on operational proxies; conservative margins applied to compensate for exposure uncertainty.
No shipment forecast for prospective cargo volume by route.	Cargo pricing is scenario-based and should be refreshed once firm shipment plans are available.
Personnel file is not split by solar system or station.	WC pricing by system is partly allocation-based rather than directly observed; allocation adds uncertainty.
Potential naming inconsistencies in equipment taxonomy.	Requires harmonisation before prediction and may introduce classification noise in frequency models.
Some variables appear outside data dictionary ranges.	Pragmatic cleaning applied; may affect model stability and tail fit at extremes if residual errors remain.
No explicit dependence structure supplied across lines or systems.	Tail capital estimates are sensitive to scenario design assumptions and should be subject to ongoing stress testing.

Table 8 - Data Limitation Summary